

Alex Hadidi

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Professional Summary

Mechanical engineer with 2+ years of design consulting experience developing robotics actuation hardware, including 13 custom actuator motor driver modules taken from concept through pilot builds. Comfortable both in the shop and behind a desk, with experience prototyping quickly, validating designs with structural and thermal analysis, documenting with GD&T, and working closely with vendors and technicians.

Education

University of California, Los Angeles (UCLA)
Bachelor of Science in Mechanical Engineering

June 2023

Cumulative GPA: 3.95/4.00

Skills

CAD/FEA: SolidWorks, Siemens NX, GrabCAD, Onshape, Fusion 360, Ansys Mechanical

Software: MATLAB, Microsoft Excel, Python, C++/Arduino, LaTeX, Teamcenter PLM, Oracle Agile PLM

Manufacturing: CNC/Manual Machining, 3D Printing, Injection Molding, Silicone Casting, Water Jetting, Laser Cutting

Engineering Experience

Fikst Product Development

Wilmington, MA

Mechanical Engineer

January 2024-Present

- Owned ground-up design for 13 custom actuator motor driver modules. Designed machined, injection molded, and waterjet components, validated thermal performance in SolidWorks Simulation, and created 30+ drawings with GD&T.
- Refined driver module prototypes using 3D printing, laser/waterjet cutting, Arduino, silicone casting, and hand tools.
- Built 20+ pilot module assemblies, identified DFA improvements, and integrated changes before handoff to technicians.
- Led redesign of 6 BLDC stator assemblies to eliminate recurring production line stoppages and performance failures. Designed machined housings, prescribed updates to stator construction, and defined changes to assembly procedure.
- Confirmed root cause elimination through tolerance stackups, hand calcs, and structural analysis in Ansys Mechanical.
- Co-led internal reviews of 6 vendor strain-wave rotary actuator designs, assessing actuator architecture, bearing arrangements, and IP54 sealing. Drove changes to motor driver integration, cable routing, and DFMA.
- Designed structures and integrated actuator joints for pick-and-place robots balancing mass, kinematics, and stiffness.
- Developed, built, and tested a robot dress pack routing vacuum, positive pressure, power, and signal to end-of-arm tool.

Apple Inc.

Cupertino, CA

iPhone Power Product Design Intern

June 2022-September 2022

- Redesigned AC power electronics assembly in Siemens NX under packaging, safety, and DFMA constraints.
- Diagnosed issues, redesigned parts, and built prototypes for a new power adapter blade mechanism.
- Reviewed vendor designs and provided feedback, ensuring design robustness and compliance with UL specifications.

Northrop Grumman Corporation

Los Angeles, CA

Systems Test Engineering Intern

June 2021-August 2021

- Ran component-level electrostatic discharge testing on FDM printed ABS parts and published results.
- Investigated printed wiring board sample contamination and developed processing procedure to allow for inspection.
- Reviewed detail and assembly engineering drawings for quality, accuracy, and ease of interpretation.

Activities & Leadership

UCLA Biomechatronics Lab

Los Angeles, CA

Undergraduate Researcher

October 2022-June 2023

- Designed and manufactured molds for casting custom elastomer end effectors for Hello Robot's Stretch platform.
- Assisted with development and manufacturing of load cell calibration setup for end effector characterization.

UCLA Bruin Racing

Los Angeles, CA

Internal Vice President

April 2021-May 2022

- Served on administrative board of 100+ member-strong organization to guide club-wide decisions.
- Managed relations and events between teams, focusing on technical knowledge transfer and professional development.

Project Engineer - Baja SAE

October 2019-May 2022

- Redesigned ball screw linear actuator for Baja SAE vehicle transmission, addressing failure mode of previous version.
- Designed and manufactured Bruin Racing Baja's first active cooling system for custom off-road vehicle transmission.
- Optimized design and analyzed airflow through vehicle transmission using SolidWorks Flow Simulation software.